வாய்ப்பு மாறி Yஇன் இறுதிநிலை நிகழ்தகவு சார்பு

$$g(y)=\begin{cases}2y, & 0 \le x \le 1 \\ 0 & \text{மற்றபடி}\end{cases} \quad \text{எனில்} \quad E(X+Y)= E(X)+E(Y)$$

தீர்வு:

$$E(X+Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x+y)f(x,y)\,dxdy$$

$$= \int_{0}^{1}\int_{0}^{1}(x+y)4xy\,dxdy$$

$$= 4\left[\int_{0}^{1}\int_{0}^{1}x^2y\,dx\,dy+\int_{0}^{1}\int_{0}^{1}xy^2\,dx\,dy\right]$$

$$= 4\left[\int_{0}^{1}\left(\int_{0}^{1}x^2\,dx\right)ydy+\int_{0}^{1}\left(\int_{0}^{1}y^2\,dy\right)xdx\right]$$

$$= 4[\int_{0}^{1}\frac{1}{3}y\,dy+\int_{0}^{1}\frac{1}{3}x\,dx]$$

$$= \frac{4}{3}\left[\int_{0}^{1}y\,dy+\int_{0}^{1}x\,dx\right]$$

$$= \frac{4}{3}[\frac{1}{2}+\frac{1}{2}]=\frac{4}{3} \qquad \ldots (1)$$

$$E(X) = \int_{-\infty}^{\infty}xf(x)\,dx$$

$$= \int_{0}^{1}x\times 2x\,dx = 2\int_{0}^{1}x^2\,dx$$

$$E(X) = 2\left[\frac{1}{3}\right]=\frac{2}{3}$$

$$\mathrm{E(Y)} = \int_{-\infty}^{\infty}yf(y)\,dy = \int_{0}^{1}y\times 2y\,dy$$

$$= 2\int_{0}^{1}y^2\,dy$$

$$E(Y) = 2\left[\frac{1}{3}\right]=\frac{2}{3}$$

$$\mathrm{E(X)}+E(Y) = \frac{2}{3}+\frac{2}{3}=\frac{4}{3} \qquad \ldots (2)$$

(1) மற்றும் (2) லிருந்து

$$E(X+Y) = E(X)+E(Y)$$

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